

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

INDEX
NUMBER

PRELIMINARY EXAMINATION 2024 SECONDARY 4 NORMAL ACADEMIC

MATHEMATICS

Paper 1

4045/01

2 hours

Marking Scheme

Question	Answer
1a	0.00691
1b	6.9084×10^{-3}

Question	Answer
2	$6xy - y^2$ or $y(6x - y)$

Question	Answer
3	$\frac{97776}{116.4} \times 100$
	$= 84000$

Question	Answer
4	$A = \sin^{-1}(0.8)$ $A = 0.927$ or 2.21
	<i>Since A is obtuse,</i> $A = 2.21$

Question	Answer
5a	$m = 1$
5b	$k = \text{any negative value}$
Question	Answer
6	$y = k\sqrt{x}$
	$0.6 = k\sqrt{9}$ OR $\frac{y}{\sqrt{100}} = \frac{0.6}{\sqrt{9}}$ $k = 0.2$
	$y = 0.2(\sqrt{100})$
	$y = 2$

Question	Answer
7	$\frac{PA}{PQ} = \frac{1}{4}$ OR $\frac{AB}{QR} = \frac{x}{x+9}$
	$\frac{x}{x+9} = \frac{1}{4}$
	$4x = x + 9$ $3x = 9$
	$x = 3$

Question	Answer
8a	6
8b	Median $= \frac{27 + 29}{2}$ $= 28$

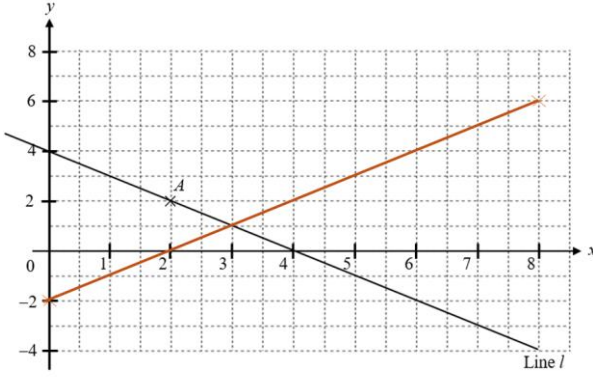
Question	Answer
9a	$\frac{2}{x+3} + \frac{1}{x^2-9}$ $= \frac{2(x-3)+1}{(x-3)(x+3)}$ $= \frac{2x-5}{(x-3)(x+3)}$
9b	$4p+1 > 15$ $4p > 14$
	$p > 3.5$
	Smallest prime number = 5

Question	Answer
10a	1cm : 250 km
	Actual Distance = 18.4×250 km
	= 46 00 km
10b	$(1\text{cm})^2 : (250\text{km})^2$
	$1\text{cm}^2 : 625\,00\text{km}^2$
	Map area of Japan $= \frac{377973}{62500}$
	= 6.05cm^2

Question	Answer
11	$\frac{2x}{3} = \frac{4}{x-5}$
	$2x(x-5) = 12$
	$2x^2 - 10x - 12 = 0$
	$x^2 - 5x - 6 = 0$
	$(x-6)(x+1) = 0$ <i>OR</i> $x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(-6)}}{2(1)}$
	$x = 6$ or -1

Question	Answer
12	Triangle ABD and Triangle BCD have the <i>same height</i>. $\frac{\text{Area of triangle } BCD}{\text{Area of triangle } ABD} = \frac{BC}{AB}$ $\text{Area of triangle } BCD = \frac{12}{3} \times \text{Area of triangle } ABD$ $\text{Area of triangle } BCD = 4 \times \text{Area of triangle } ABD$ OR Let x be the height of both triangles. $\text{Area of triangle } ABD = \frac{1}{2}(3)(x) = 1.5x$ $\text{Area of triangle } BCD = \frac{1}{2}(12)(x) = 6x = 4(1.5x)$ OR state that “<i>the ratio of the area of triangle BCD and triangle ABD is equal to the ratio of their bases.</i>” Sam is correct.

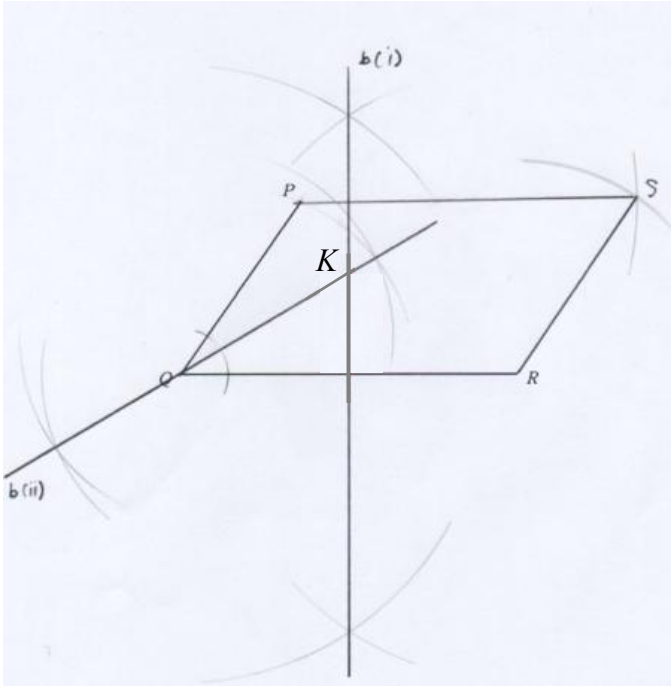
Question	Answer	Marks
13a	Acceleration $= \frac{20}{25}$	2
	$= 0.8$	
13b	Total distance travelled $= \frac{1}{2}(25 + 60)(20)$	2
	$= 850 \text{ m}$	

Question	Answer	Marks
14a	$\frac{10 - 2}{n - 2} = -\frac{4}{4}$	1
	$8 = -(n - 2)$	
	$n - 2 = -8$	
	$n = -6$	
14b(i)		1
14b(ii)	(3,1)	1

Question	Answer
15a	$a = 5$
	$k = 25$
15b	$(x+5)^2 = 8$
	$x = \pm\sqrt{8} - 5$
	$x = -2.17 \text{ or } -7.83$

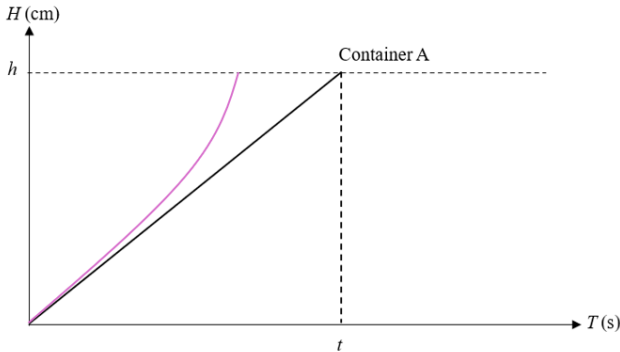
Question	Answer
16a	7:4
16b	3 units represents 6cm
	11 units represents 22cm
	$AB = 22 \text{ cm}$

Question	Answer
18	$8^2 = 3^2 + r^2$
	$r = \sqrt{55}$
	Surface Area $= \pi(\sqrt{55})^2$
	$= 173 \text{ cm}^2 \text{ (3 sig fig)}$

Question	Answer
19a	
19b	(Refer to the above) Perpendicular bisector of QR constructed
	bisector of angle PQR
19c	marked point of intersection with K

Question	Answer
20	$\frac{149}{360} \times 2\pi r = 18.2$
	$r = 6.9985$
	Area of minor sector $= \frac{149}{360} \times \pi(6.9985^2)$ $= 63.6867$
	Area of triangle $= \frac{1}{2}(6.9985)(6.9985)\sin 149^\circ$ $= 12.613$
	Area of minor segment $= \text{area of minor sector} - \text{area of triangle}$ $= 63.6867 - 12.613$ $= 51.0737$
	Area of major segment $= \pi(6.9985)^2 - 51.0737$
	$= 103 \text{ cm}^2$

Question	Answer	
21	$3x + 8y = 11 \quad \text{-----} \quad \textcircled{1}$ $4y = 3 - x \quad \text{-----} \quad \textcircled{2}$ $\textcircled{2} \times 2 : \quad 8y = 6 - 2x \quad \text{-----} \quad \textcircled{3}$ Sub $\textcircled{3}$ into $\textcircled{1}$ $3x + 6 - 2x = 11$ OR $x = 3 - 4y \quad \text{-----} \quad \textcircled{3}$ Sub $\textcircled{3}$ into $\textcircled{1}$ $3(3 - 4y) + 8y = 11$	
	$x = 5$	$y = -\frac{1}{2}$
	$y = -\frac{1}{2}$	$x = 5$

Question	Answer
22	 Graph sketched is an increasing non-linear function starting from origin. Graph sketched is above the graph of Container A and end at the height of h cm

Question	Answer
23a	Sam may have seen that the <u>sectors for representing the T-shirts ordered for both years are the same</u> (90°), the number of S-sized T-shirts ordered may not be the same because the total number of Orientation T-shirts ordered may not be the same for 2022 and 2023.
23b	Angle of the sector that represent M -sized shirts for year 2023 = 75° Number of M-sized T-shirts $= \frac{240}{360} \times 75$ $= 50$